

FairWeight: Private Fairness-aware Aggregation using Model Interpretation based Weightage in Federated Learning

Harsh Kasyap*, Ugur Ilker Atmaca[†], Carsten Maple[‡], Nic Lane[§]

Abstract—Federated learning (FL) has been adopted as a promising solution for collaborative privacy-preserving machine learning. FL provides independence over the training data, but this can lead to statistical heterogeneity, which may result in issues such as bias and class imbalance. Thus, a global model, even with high accuracy, may be prone to discrimination and perform well only for certain sensitive attributes and classes. Global accuracy is not a perfect measure of the model's effectiveness; it should be reported in terms of balanced accuracy and fairness. Existing solutions attempt to mitigate bias using local or global debiasing techniques. However, local debiasing results in poor performance with increasing heterogeneity, and global debiasing is either designed only for achieving fairness, ignoring balanced accuracy, or costly in terms of finding an optimal solution.

We present a novel model interpretation based bias mitigation aggregation scheme in FL, called FairWeight. It calculates the aggregation weightage, by considering both fairness and loss for biased samples from the minority class. After optimizing the combined loss for prediction and fairness, the important weights responsible for bias and misclassification are calculated locally between clients. These weight indices are shared using secure aggregation to calculate the final aggregation weightage. Further, a secure aggregation scheme is devised that penalises biased weights across clients, rather than punishing the global model. FairWeight is evaluated on five real-world datasets and compared against five state-of-the-art fairness-aware aggregation schemes. The results provide empirical evidence that FairWeight achieves both balanced accuracy and fairness, with reduced computation and communication overheads.

Index Terms—Federated Learning, Bias, Fairness, Model Interpretation, Privacy, Secure Aggregation

I. INTRODUCTION

Federated learning (FL) enables collaborative model training across distributed clients without sharing raw data, thus offering a promising approach for privacy-preserving machine learning [1], [2]. FL restricts the movement of data, by bringing the model to the edge and allowing it to train locally. Only the local model is sent back to the server for aggregation. FL has been adopted across domains, including sensitive applications such as healthcare [3], [4] and finance [5], [6], [7]. However, FL is increasingly challenged by fairness issues,

particularly when client data distributions are heterogeneous and reflect social or institutional biases.

Data heterogeneity in FL often extends beyond label imbalance to include sensitive or protected attributes such as gender, race, or geographic origin [8]. For instance, in a nationwide loan risk prediction model, regional banks may differ in customer demographics, some may predominantly serve male clients, others may have skewed loan approval histories. It is possible that, in a particular region, 90% of the bank customers are male and have only 10% records for failure in loan repayment. Such inherited biases may reinforce existing prejudices and amplify existing societal inequities, possibly harming vulnerable communities [9]. Notably, the Federal Reserve Board, USA [10], reveals that though racial discrimination in mortgage lending has declined compared to 30 years before, it is still found that 17% of black borrowers are denied loans compared to only 8% of whites. Thus, collaboration improves the overall model performance but also propagates bias in the global model toward a particular group of sensitive attributes [11] or class [12]. Such statistics underline the potential harm of unfair decision making in FL and motivate the need for fairness-aware aggregation [11], [13].

Existing approaches to mitigate biases in FL can be classified as local [14], [15], [16] or global [17], [12] debiasing techniques. Local debiasing operates independently at each client [14], [15], [16], making it communication-efficient but less effective in the presence of highly heterogeneous data [18], [19]. In contrast, global debiasing techniques [17], [12] coordinate across clients to reduce systemic bias but often require costly communication or privacy trade-offs. For example, FairFed [17] requires fairness evaluation at both local and global levels. In another study, FairTrade [12] empirically showed that state-of-the-art debiasing techniques minimize discrimination at the cost of balanced accuracy. Suppose a binary classification model always predicts positive results for both classes. Given 90% samples from the positive class and 10% from the negative class, accuracy is 90%, demographic parity is 0 (as the model predicts 1 irrespective of sensitive attribute), but balanced accuracy is 50%. FairTrade utilises a multi-objective optimization to jointly maximize balanced accuracy and minimize bias, though at significant computational and communication cost.

To address the above-discussed concerns, we propose a novel aggregation weightage scheme for fairness-aware Aggregation called FairWeight. While the existing techniques

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incentivise or penalise the complete model for bias, FairWeight is the first attempt to perform coordinate-wise fairness-aware aggregation in FL. First, the clients train locally, optimizing loss for both accuracy and fairness, using fairness constrained optimization [20]. Then, they identify the weights responsible for bias and misclassification for minority class samples. For this, FairWeight uses Shapley value using random sampling to calculate weights' importance, with data from all groups of the sensitive attribute. Then, after establishing privacy-preserving communication among clients, the average aggregation weightage score is calculated coordinate-wise, for each local model update. FairWeight performs adaptive aggregation, which penalises selectively, and the weightage score is assigned on the basis of normalised frequency of responsible weights. FairWeight has been evaluated over multiple datasets and compared with state-of-the-art fairness-aware aggregation schemes in FL. It achieves both high balanced accuracy and low discrimination score, with reduced computation and communication costs. Summarising the key contributions, we highlight, that FairWeight achieves the following benefits over existing state-of-the-art debiasing techniques:

- **Coordinate-wise fair aggregation weightage score.** FairWeight identifies the responsible weights causing bias and misclassification across clients, after performing local fairness-constrained optimization. It assigns an average aggregation weightage score coordinate-wise, based on normalised frequency of responsible weights. It achieves both balanced accuracy and fairness.
- **Private and communication-efficient aggregation.** FairWeight can be tailored with either of secure aggregation (SecAgg) [21] or homomorphic encryption (HE) [22], to achieve privacy-preserving computation. HE-based solution reduces the number of interactions between the client and the server from two in FairFed [17] and many in FairTrade [12] to one in FairWeight.

Organisation. The remainder of this paper is organized as follows. Section II discusses the background of federated learning, fairness metrics and fairness-aware federated learning aggregation. Section III discusses the adversarial settings and desired security guarantees. Section IV presents FairWeight, a model interpretation based weightage for fairness-aware aggregation. Section V describes the simulation setup, experiment, and results. Section VI concludes this work.

II. BACKGROUND AND RELATED WORK

A. Federated Learning

Federated Learning (FL) is a privacy-preserving collaborative machine learning technique [1], where model training is decentralized across multiple clients. Consider a scenario with n clients $C_1, C_2, \dots, C_n$ collaborating with a central server S to jointly learn a global model represented by parameter vector θ . The goal is to minimize the weighted average of the local loss functions across all participating clients, formulated as:

$$\min_{\theta} f(\theta) = \sum_{k=1}^n w_k L_k(\theta). \quad (1)$$

Here, $L_k(\theta)$ denotes the local objective function of the client C_k , and $w_k \geq 0$ is a weight coefficient proportional to the size of the local dataset D_k , such that $\sum_{k=1}^K w_k = 1$. The local objective is defined as the empirical risk over the local dataset, i.e.,

$$L_k(\theta) = \frac{1}{|D_k|} \sum_{(x,y) \in D_k} \ell(\theta, x, y), \quad (2)$$

where $\ell(\cdot)$ is the loss function evaluated on input output pairs (x, y) from D_k .

In the standard Federated Averaging (FedAvg) algorithm [1], the raw local model updates are shared with the server are prone to inference attacks [23], [24]. To enhance privacy, secure aggregation [21] is often employed in FL. This cryptographic protocol allows the server to compute the sum of the model updates from clients without learning any individual client's update. That is, each client secret shares its update, and only the aggregated model is revealed to the server, preserving the privacy of local training data.

FL can also suffer from performance disparities. Specifically, the global model may exhibit demographic bias due to statistical heterogeneity (non-IID nature) in client datasets [17], and poor balanced accuracy in classification tasks due to class imbalance across clients [12]. These challenges motivate the development of private fairness-aware FL aggregation.

B. Fairness

A fair model is generally described as one that does not demonstrate any bias or favouritism towards an individual or a group due to their inherent or acquired characteristics. Bias can impact the model's fairness at both the individual and group levels. Individual fairness requires that the model make similar predictions for individuals with similar characteristics, whereas group fairness ensures equitable treatment (in the aggregate) across different groups defined by sensitive attributes, such as race, gender, or age. We consider achieving group fairness in a binary classification problem where clients have datasets having a single sensitive attribute (A) with binary values. However, this can be extended to multi-class attributes, employing a one-vs-all approach. For example, consider a dataset has gender and race as sensitive attributes. For this evaluation, (1) consider employing our algorithm first on the basis of gender, (2) update the model and (3) repeat the same process on the basis of race. Similar methodology has been adopted in [25] and it has been observed that, this significantly reduced the bias metrics on the basis of both the sensitive attributes.

Fairness Metrics. Y represents the true value, and $\hat{Y}$ signifies the predicted result. Fairness can be measured using different metrics, such as demographic parity, equalised odds, causal fairness, global fairness and balanced accuracy.

Definition 1 (Demographic Parity). *Demographic parity (i.e., Statistical parity) ensures an equal probability of a positive outcome across different groups defined by a sensitive attribute [26]. Formally, it is defined as:*

$$\left| P(\hat{Y} = 1 \mid A = 0) - P(\hat{Y} = 1 \mid A = 1) \right| \leq \epsilon. \quad (3)$$

A smaller value ϵ signifies a higher degree of fairness, and the statistical parity is satisfied if ϵ is 0. It ensures that the model does not favour one group over another in terms of the likelihood of positive outcomes.

Definition 2 (Causal Group Fairness). *Causal fairness involves analysing the causal structures in the data to decouple predictions from sensitive attributes. The causal group fairness notion average treatment effect (FACE) [27] is a commonly used metric to assess the discrimination embedded in the predictions. It is further modified to consider predicted outcomes in [12] as:*

$$\begin{aligned} \text{FACE} &= E \left(\left| Y_{\text{pot}}^{A_1} - Y_{\text{pred}}^{A_1} \right| - \left| Y_{\text{pot}}^{A_0} - Y_{\text{pred}}^{A_0} \right| \right) \\ &= \frac{1}{n} \sum_{i=1}^n \left| Y_{\text{pot}_i}^{A_1} - Y_{\text{pred}_i}^{A_1} \right| - \left| Y_{\text{pot}_i}^{A_0} - Y_{\text{pred}_i}^{A_0} \right|, \end{aligned} \quad (4)$$

where $Y_{\text{pot}}^{A_1}$ and $Y_{\text{pred}}^{A_1}$ represent the potential and predicted outcomes when $A = 1$. The difference in the true positive outcomes between the protected and non-protected groups is measured. A value of 1 or -1 indicates complete unfairness, while 0 means complete fairness. Unlike traditional statistical measures, it aims to prevent sensitive attributes from indirectly influencing the predictions through other variables.

Definition 3 (Fairness Constrained optimization). *The task is to train a predictive model at the client, satisfying fairness constraints [20]. It can be defined as:*

$$\underset{\theta}{\text{minimize}} \quad L_k(\theta) + \zeta \times F(\theta) \text{ s.t. } g(\theta) \leq e, \quad (5)$$

where $L_k(\theta)$ is the local predictive loss function and $F(\theta)$ is the fairness loss, with given regularisation constraint ζ . $g(\theta)$ is the fairness metric with given budget e . For each fairness metric, a set of linear constraints is generated in the form of $Q\eta(f) \leq e$, where Q is a matrix and $\eta(f)$ is a vector with conditional moments. The authors in [12] define a constraint for Demographic Parity (DP) as:

$$F(\theta) = \zeta \times \|(\text{ReLU}(Q\eta(f))) - e\|_2. \quad (6)$$

In the context of binary classification with a binary sensitive attribute, DP can be represented using a pair of equality constraints. Specifically, for each group A , we define a positive deviation constraint $\Delta^+ := \eta_A(f) - \eta_{\{X \setminus A\}}(f) \leq 0$ and a negative deviation constraint $\Delta^- := -\eta_A(f) + \eta_{\{X \setminus A\}}(f) \leq 0$. These constraints ensure that the conditional moment $\eta_A(f)$, which often represents the average prediction output for the group A , remains close to that of the complementary group. The matrix Q is constructed to encode these constraints as follows:

$$Q_{(A, \Delta^+), A'} = \begin{cases} 1 & \text{if } A' = A \\ -1 & \text{otherwise} \end{cases}. \quad (7)$$

$$Q_{(A, \Delta^-), A'} = \begin{cases} -1 & \text{if } A' = A \\ 1 & \text{otherwise} \end{cases}. \quad (8)$$

Here, $Q_{(A, \Delta^+), A'}$ captures the degree to which the group's average prediction exceeds that of the rest of the population, enforcing an upper bound on this disparity. Conversely, $Q_{(A, \Delta^-), A'}$ captures the lower bound by reversing the sign structure, ensuring that the group's average prediction is not substantially lower than that of others. Together, these constraints maintain DP by bounding deviations in both directions.

Similarly, constraints for FACE can be defined.

Definition 4 (Global Fairness). *Data is heterogeneously distributed across clients in FL, which requires another fairness level, i.e., Global fairness. It aims to ensure that the global model does not favour any particular demographic group across the entire distributed dataset. We define it using the example of demographic parity.*

$$\sum_{k=1}^K w_k \left| P(\hat{Y} = 1 \mid A = 0, D_k) - P(\hat{Y} = 1 \mid A = 1, D_k) \right|. \quad (9)$$

It can be computed at the server using secure aggregation [17]. Similarly, balanced accuracy can be defined as $\sum_{k=1}^K w_k BA_k$, where w_k is the weightage score defined based on the proportion of data samples held with the client k with respect to the total data samples among the clients and BA_k is the local balanced accuracy [12].

C. Fairness-aware Federated Learning.

FairFed [17] is a fairness-aware aggregation scheme, which is server-agnostic and supports any local debiasing scheme. It finds the gap in the fairness metric from global fairness, to calculate the aggregation weightage. It performs better in highly heterogeneous data distributions. If all clients have identical (or very similar) local fairness metrics, it defaults to the FedAvg-style weighting. FairFed also supports fairness-adjusted aggregation, using the parameter β , which controls the influence of the fairness mismatch. FairTrade [12] empirically shows that existing schemes only achieve fairness at the cost of balanced accuracy. They applied fairness-constrained optimization locally and further utilised multi-objective optimization to find a balance between balanced accuracy and fairness. Since FairFed and FairTrade involve communication with the server, they utilise SecAgg to ensure no privacy leakage. However, they incur heavy computation and communication overheads. Agnostic-Fair [28] adds regularisation terms to reweight training samples to mitigate bias. FCFL [29] provides a consistent Pareto utility (accuracy and fairness) distribution, using a gradient-based approach. FedFB [30] adapts the centralised Fair-Batch [31] to locally mitigate bias in a federated setting. It employs a two-level optimisation methodology in which the server uses secure aggregation to compute group-specific loss statistics, which are then used to update global reweighting coefficients for sensitive groups. These coefficients are returned to clients so they can adjust their local loss functions. However, these schemes punish the local models based on less information and result in poor performance in terms of balanced accuracy and global group fairness.

We also establish a clear distinction between FairWeight and prior aggregation schemes based on coordinate-based aggregation algorithms, such as Trimmed Mean [32] and FLTC [33]. They primarily focus on mitigating poisoning attacks, without explicitly reasoning about fairness. Similarly, SHAP-based aggregation methods, including FaShapley, leverage model interpretation to assess client contributions and robustness, but their primary objective remains adversarial robustness rather than fairness guarantees. While some recent works, such as FedVal [34], incorporate fairness considerations, their notion of fairness is different. In particular, they address fairness in terms of minority or underrepresented labels rather than sensitive attributes (e.g., gender, race, or age). Moreover, existing fairness-aware algorithms discussed above, penalize the complete model. In contrast, FairWeight is explicitly designed to integrate fairness-aware weighting into the aggregation process by considering sensitive attributes alongside robustness.

Further, Shaham et al. [35] provide a taxonomy for the relationship between privacy and fairness, dividing it into “Aligned” and “Contrasting” views. It is noted that Differential Privacy (DP) may cause a drop in fairness because the added noise impacts minority groups more severely, as a privacy-fairness trade-off. FairWeight avoids such a fairness drop by using SecAgg/HE rather than introducing noise. The privacy-fairness trade-off is also discussed in [36], as improving individual fairness by treating similar nodes may inherently increase privacy risks in order to perform fairer predictions. The authors discussed edge inference attacks in Graph Neural Networks (GNNs). To mitigate this, the authors propose a fine-tuning method that balances the conflicting goals of fairness and privacy. Although Fairweight aims to achieve utility, fairness, and privacy altogether.

III. THREAT MODEL

We consider an FL setting in which multiple clients collaboratively train a global model under the orchestration of a server, without sharing their raw data. Our threat model focuses on the privacy and fairness aspects of FL in the presence of data heterogeneity and potential adversarial behavior.

Adversarial Setting. We assume the presence of *honest-but-curious* (*semi-honest*) clients and an *honest-but-curious* server. All parties correctly follow the protocol, but may try to infer sensitive information from the messages they receive. The adversary’s objective is (1) to infer individual client data from their local model updates or global model update; (2) to identify sensitive attribute distributions or fairness violations among clients.

Security Guarantees. The goal is to ensure fairness in the aggregated model despite potential data biases, while protecting the privacy of each client’s local data.

IV. FAIRWEIGHT: MODEL INTERPRETATION BASED WEIGHTAGE FOR FAIRNESS-AWARE AGGREGATION

This section presents the proposed fairness-aware aggregation scheme in FL, called FairWeight. It is devised based on fairness-constrained optimization and a model interpretation technique for identifying the responsible weights for bias and

misclassification. Rather than allocating the same weightage to the complete model, it assigns the aggregation weights coordinate-wise, based on their frequency across clients. In this way, it achieves both balanced accuracy and group fairness. Moreover, FairWeight facilitates privacy-preserving computation, ensuring that there is no privacy leakage for both the local model and biased weights. Table I lists the summary of notations used.

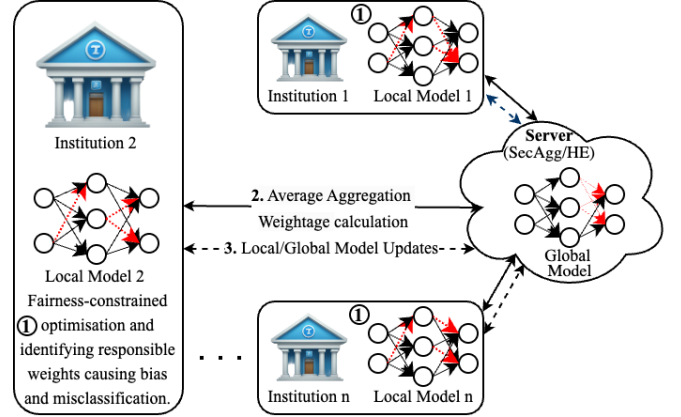


Fig. 1: FairWeight: Workflow diagram.

TABLE I: Summary of Notations Used.

Symbol	Meaning
n	Total number of clients in federated learning
i, j	Client index ($i \in \{1, \dots, n\}$), Weight index
t, T	Iteration $t \in T$, total T iterations (rounds)
$D_i, D_i $	Local dataset of client i , Size of local dataset D_i
θ	Global model parameters
θ_i	Local model parameters of client i
θ_i^t	Local model parameters of client i in round t
w_i	Weight coefficient for client i
w_{ij}	Weight coefficient j for client i
w_i^t	Weight coefficient for client i in round t
A	Sensitive (protected) attribute
s, s'	Two groups of sensitive attribute A (e.g., male/female)
d, d'	Datasets associated with sensitive attribute values s, s'
ϕ_i	Shapley value contribution of weight i
WI	Weight importance score (via Shapley value)
freq_j	Frequency of weight j marked as biased across clients
β_1, β_2	Hyperparameters for incentivizing/penalizing weights
SecAgg	Secure Aggregation protocol
HE	Homomorphic Encryption scheme

Overview. FairWeight follows a modular architecture with primarily four components: (i) a *local fairness-constrained training module*, (ii) *weights importance estimator*, (iii) *weightage importance estimator*, and (iv) a *privacy-preserving secure aggregation module*. Each module performs a well-defined role in the end-to-end aggregation process. Figure 1 illustrates the interaction among these modules. The modules can be briefly described as follows: (1) All the participating institutions (clients) train over their locally held dataset, using the previous (or preliminary) global model. They perform fairness-constrained optimization, which considers minimizing both training and fairness loss. (2) Then, clients use the model interpretation technique (employing Shapley value) to identify the responsible model weights causing bias and misclassification, using the data from both groups of the sensitive attribute.

The weights highlighted in dotted red represent the responsible weights, across the clients. (3) Instead of revealing raw weight indices, all clients contribute the encrypted indices of the most responsible weights to a secure aggregation protocol (such as SecAgg or HE) to calculate the frequency of such weights across the clients. It assists in calculating the individual aggregation weightage. Following that, they calculate the sum of all the individual aggregation weightage, which further assists in calculating the final average aggregation weightage. (4) This operation is executed privately using either SecAgg or HE, without leaking any information to other participants or the server. Finally, clients do secure aggregation to calculate the global model and global fairness metrics, achieving both balanced accuracy and fairness. Now, we discuss each step in detail.

1) *Local model training*: Each participating client (k) owns its private dataset (D_k), which can have a biased distribution in terms of class or sensitive attributes. While training, the local model (θ_k) also learns to discriminate on the basis of sensitive attributes or performs poorly due to class imbalance. Then, the client performs fairness constrained optimization locally on θ_k , using any fairness metric of interest. For example, clients can perform optimization as defined in Equation 5 (Definition 3), to minimise both the prediction and demographic parity loss. However, this does not guarantee that the aggregation of these local models will result in a fair global model that achieves high balanced accuracy and fairness. This is because, the aggregation weightage scores assigned to these models are proportional to their data samples [37] or based on fairness loss [17]. FairWeight further identifies the weights contributing to the local bias and misclassification.

2) *Identifying responsible weights*: The ability to identify the most significant weights contributing to bias and misclassification in the model stems from the model's own capability to differentiate based on the protected attribute. For instance, a loan risk prediction (binary) model to assess the risk of approving a loan should base its predictions on significant features such as credit score, salary, and other relevant factors. However, it also learns to discriminate on the basis of other sensitive attributes such as gender, race, etc. Thus, the intuition is to identify the weights that contribute the most to causing this discrimination. Suppose x_s is the sensitive attribute, with two groups s and s' . If x_s has more than two values, a one-to-all approach can be used to categorize them as s and s' . Assume D_k has the majority of positive (*pos*) samples, and very few negative (*neg*) samples, especially for s' . Since the classification problem is two-fold: (1) most of the samples with s' are classified as *neg* due to bias; (2) mitigating the bias causes true *neg* samples with s' also classified as *pos*.

The intuition is to first determine the weights importance (WI) for: (i) tp_s : *pos* samples with s , also classified as *pos*; (ii) $fn_{s'}$: *pos* samples with s' , but classified as *neg*; (iii) $tp_{s'}$: *pos* samples with s' , also classified as *pos*; and (iv) $fp_{s'}$: *neg* samples with s' , but classified as *pos*. The bias is primarily dominated by the samples from (i) and (ii), while the misclassification is dominated by (iii) and (iv). Then, adjusting the weights with the highest WIs of (i-ii) mitigates

bias and (iii-iv) mitigates bias and misclassification. These combinations should be changed based on the direction of bias changes. FairWeight follows a two-step process: (1a) calculates the WI for a given model (f), with respect to both datasets of both sensitive attributes (say d and d'); (1b) selects the weights with the highest differences between WIs of d and d' .

2a. To calculate the weights importance (WI), FairWeight uses a model interpretation technique employing Shapley value (SV). SV determines the importance of all participants (weights) involved in the prediction task as follows [38]:

$$\phi_j(v) = \sum_{S \subseteq W \setminus \{j\}} \frac{|S|!(w - |S| - 1)!}{w!} (v(S \cup \{j\}) - v(S)), \quad (10)$$

where $\phi_j(v)$ represents the importance of weight j , given $W = \{1, \dots, w\}$ weights. This importance is calculated by determining the marginal contribution of each weight with all possible combinations. S represents all possible subsets of weights. The characteristic function is denoted by v , with $v(S)$ being the prediction for the weights in subset S , marginalized over the weights not included in S . This is computed through multiple integrations as follows:

$$v(S) = \int \hat{f}(x_1, \dots, x_w) dW_{x \notin S} - E_X(\hat{f}(X)). \quad (11)$$

Since this is computationally intensive, with the computational complexity of $\mathcal{O}(2^w)$, given w weights. This can be calculated by repeatedly sampling k weight parameters from a total of w . By applying unstructured pruning of these k weights, using explanations provided by SV, this becomes highly time-efficient, reducing the computational complexity from exponential to linear ($\mathcal{O}(k)$).

The marginal contribution of weights ($v(S \cup \{j\}) - v(S)$) in Equation 10 can be calculated as the absolute value of the product of gradients and weights [39], as:

$$\begin{aligned} |v(S \cup j) - v(S)| &= |-L(\theta + d\theta_j, x, y) + L(\theta, x, y)| \\ &\approx \left| \frac{\partial L}{\partial \theta_j} d\theta_j \right| = \left| \frac{\partial L}{\partial \theta_j} \right| |d\theta_j| = \left| \frac{\partial L}{\partial \theta_j} \right| |\theta_j|. \end{aligned} \quad (12)$$

It is approximated by considering the first-order terms in the Taylor expansion of the cost function L . $|d\theta_j|$ can be expressed as $|\theta_j - 0|$ i.e., $|\theta_j|$. Consequently, the marginal contribution (or WI) of the weight parameters can be determined by evaluating the model and then performing backpropagation. The algorithmic details for finding WI is provided in the Appendix A.

2b. FairWeight identifies the responsible weights with the widest WI gap [25], due to different groups (d and d') from the sensitive attribute, as shown in Figure 2. A union of the responsible weights indices is taken for further scoring. We consider these weights responsible for bias and misclassification, for two groups of a sensitive attribute.

Algorithm 1 FairWeight: Model Interpretation based Weightage for Fairness-aware Aggregation

Input: Initialise θ^t (global model, $t = 0$) and θ_i^0 as θ^0 (as local model updates $\forall i \in n$), x_s (sensitive attribute with s and s' as two groups), Local Data Statistics ($ds_i^t = Acc_i^t, \{Pr(A=1, Y=1), Pr(A=0, Y=1)\}$) from clients for calculating global fairness metrics using SecAgg. Initialise $1 \leq \beta_1 < \frac{n}{n-1}$ and $\beta_2 \geq 1$, and adjust based on application sensitivity.

Output: θ^{t+1} : global model for next iteration, balanced accuracy (BA_g) and global fairness metrics ($Fair_g$).

- 1: **for** each round t in $1, 2, \dots, T$ **do**
- 2: each participant ($i \in n$) trains θ_i^t , and performs fairness constrained optimization locally. ▷ Definition 3.
- 3: each participant calculates w_i^t , by identifying the responsible model weights causing bias and misclassification over x_s , employing model interpretation. ▷ As shown in Figure 2.
- 4: $freq^t \leftarrow SecAgg(w_i^t)$. ▷ Clients engage in secure aggregation to calculate the frequency of weight parameters causing bias, across clients.
- 5: $w_i^t \leftarrow (\beta_1 \cdot \frac{n - (freq_j^t \cdot w_{ij}^t)}{n}) \beta_2$. ▷ Clients update the aggregation weightage score locally.
- 6: $w_{sum}^t \leftarrow SecAgg(w_i^t)$. ▷ Clients engage in secure aggregation to calculate the sum of aggregation weightage scores.
- 7: $\bar{w}_i^t \leftarrow \frac{w_i^t}{w_{sum}^t}$. ▷ Clients calculate the average aggregation weightage score.
- 8: $\theta^{t+1}, BA_g, Fair_g \leftarrow SecAgg(w_i^t \cdot \theta_i^t, ds_i^t)$. ▷ Clients engage in secure aggregation to calculate the global model, balanced accuracy, and global fairness for the next iteration.
- 9: **end for**

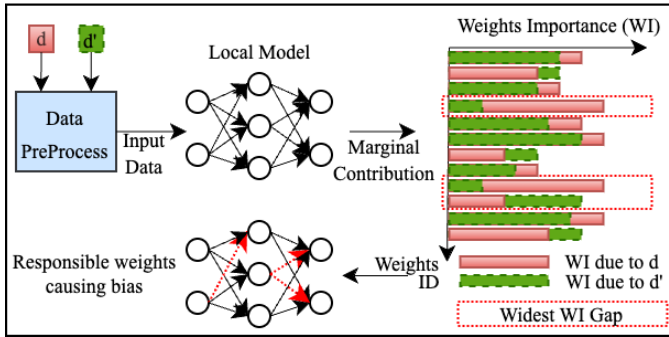


Fig. 2: Identifying responsible weights (with the widest WI gap) causing bias.

3) *Aggregation Weightage Calculation:* After identifying the responsible weights causing bias and misclassification locally, the participants have to communicate among themselves to calculate the aggregation weightage for each local model update. This operation can be executed either, using secure aggregation (SecAgg) [21] or homomorphic encryption (HE) [22]. It can be decided based on the tradeoff between computation and communication costs.

First, we discuss the steps involved in calculating the aggregation weightage, in plaintext. The weightage score is calculated coordinate-wise, such that each of the weights is incentivized or penalized based on the corresponding weights' performance in other models rather than penalizing the complete model. This calculation is also a two-step process: (2a) individual aggregation weightage calculation; (2b) average aggregation weightage calculation.

We present it with an example. Let there be three participants, with each model having four weight parameters. After executing step 1, participants have already identified the weights indices responsible for bias and misclassification.

$$w_1 = [0, 0, 0, 1], w_2 = [0, 1, 0, 1] \text{ and } w_3 = [0, 1, 1, 1],$$

where w_i stores the weights score for the model θ_i and 1 represents that the weight parameter has been identified as responsible for bias.

3a. The idea is to penalize a weight strictly if it is responsible for causing bias in more local models. Thus, we need to count the frequency of weight parameters across clients, causing bias. Frequency here denotes repeated marks of the same weight by multiple clients. Thus, it can be calculated as sum of w_i , i.e., $freq = [0, 2, 1, 3]$. A frequency closer to the number of clients signifies that the weight is causing bias in the majority of the models and should be punished strictly and vice versa. Here, weight parameter 1 has frequency zero, thus it need not be penalised across the clients. Its aggregation weightage score will be 1. While it can be observed that weight parameter 2 is causing bias in models 2 and 3, but not in model 1. So, it will be more incentivized in model 1 and penalized strictly in 2 and 3. Weight parameter 3 is causing bias only in model 3, but not in models 1 and 2. So, it will be incentivised in models 1 and 2 and penalised in 3. Weight parameter 4 is causing bias in all the models, so it will be pruned, and assigned a value of 0. Thus, the aggregation weightage scoring can be written as:

$$w_{ij} = \frac{n - (freq_j \cdot w_{ij})}{n},$$

where i represents the client, j is the position of the weight parameter in the model i , and n is the total number of participating clients. Thus, the individual aggregation weightage for the three clients can now be written as:

$$w_1 = [1, 1, 1, 0], w_2 = [1, .33, 1, 0], w_3 = [1, .33, .66, 0].$$

It can be observed that, w_{32} has score 0.33, while w_{33} has score 0.66. This is because weight parameter 2 is responsible for bias in 2 models, while weight parameter 3 is only causing bias in its own model, so it does not need to be penalised strictly. FairWeight also introduces two hyperparameters β_1 and β_2 to incentivise the honest weight parameters and strictly

penalise the responsible ones. They can be adjusted according to the application sensitivity and bias level. Thus, w_{ij} can be updated as below:

$$w_{ij} = (\beta_1 \cdot \frac{n - (freq_j \cdot w_{ij})}{n})^{\beta_2},$$

where β_1 can be any real number in $[1, \frac{n}{n-1})$ because a larger number will assign a weightage score greater than 1 to a weight causing bias. β_2 can be any real number greater than or equal to 1. They help decrease the weightage score for the responsible weights for bias and increase the remaining ones. The algorithmic details for optimizing β_1 and β_2 are provided in Appendix B.

3b. The average aggregation weightage score, for each client, $i \in n$ can be calculated as:

$$w_i = \frac{w_i}{\sum_{k=1}^n w_k}.$$

Thus, the average aggregation weightage for the three clients (keeping the hyperparameters β_1 and β_2 as 1, becomes as follows:

$$w_1 = [.33, .6, .37, 0], w_2 = [.33, .19, .37, 0], \\ w_3 = [.33, .19, .24, 0].$$

4) *Privacy-preserving Aggregation:* To calculate both the frequency and average aggregation weightage score, individual client scores need to be exchanged. To do this in a privacy-preserving fashion, we can integrate either SecAgg or HE. In Algorithm 1, FairWeight uses SecAgg to calculate the aggregation weightage scores privately. It can be observed that two SecAggs are required to calculate the weightage score, and one is required for aggregation and global fairness metrics. This is similar to the number of SecAggs required in [17].

FairWeight can also be tailored with HE, which can replace the two SecAggs required to calculate the weightage score into a single communication. Since all the operations involved in steps 4-6 in Algorithm 1 involve only addition and multiplication. Therefore, it is easily possible to apply HE. We detail the operations on the server and client sides for integrating HE in FairWeight.

Server-side. After collecting the encrypted local weights $E(w_i^t)$ from each client $i \in \{1, \dots, n\}$, the server performs the following homomorphic computations:

$$E(freq^t) \leftarrow \text{HEAdd}(E(w_1^t), \dots, E(w_n^t)), \quad (13)$$

$$E(w_i^t) \leftarrow \left(\beta_1 \cdot \frac{n - \text{HEMult}(E(freq^t), E(w_i^t))}{n} \right)^{\beta_2}, \quad (14)$$

$$E(w_{\text{sum}}^t) \leftarrow \text{HEAdd}(E(w_1^t), \dots, E(w_n^t)). \quad (15)$$

Here, HEAdd denotes element-wise addition of encrypted vectors, and HEMult represents homomorphic multiplication, which approximates the element-wise product under encryption. More details on HE, its operations and security can be referred to in the Appendix C. These operations are carried out without decrypting the data, leveraging the CKKS scheme's

support for approximate arithmetic over ciphertexts [22]. Specifically, the server aggregates to compute an encrypted frequency vector $E(freq^t)$, and computes the sum of the updated encrypted weights $E(w_{\text{sum}}^t)$, after performing required operations.

The server then transmits the updated encrypted weightage $E(w_{\text{sum}}^t)$ to all clients. These steps maintain encryption throughout the process, ensuring that the server never accesses any plaintext data.

Client-side. After receiving $E(w_{\text{sum}}^t)$, each client decrypts and calculates the average aggregation weightage score $(\frac{w_i}{w_{\text{sum}}})$, similar to step 7 in Algorithm 1. However, using SecAgg or HE is decided on the basis of a trade-off between communication and computation, depending upon the capabilities of the participants involved.

Global model and fairness metrics. Finally, clients calculate the weighted model using the average aggregation weightage score. Then, they engage in secure aggregation to calculate the global model, balanced accuracy, and fairness metrics for the next iteration. The complete procedure of FairWeight, is shown in the Algorithm 1.

V. EXPERIMENTAL EVALUATION

Datasets. FairWeight has been evaluated on five datasets: Adult, Bank, Default, KDD [40] and Law [41], under varying numbers of attributes, instances, sensitive attributes, and class imbalance ratios. The positive to negative class ratios for Adult, Bank, Default, Law, and KDD are 1:3.0, 1:7.87, 1:3.52, 1:3.50 and 1:15.11, respectively. Datasets are distributed randomly or based on specific attributes (age for Adult, Bank, Default, and KDD; income for Law) to reflect more realistic situations (to ensure the presence of non-iid data).

The datasets are discussed in detail below.

1) **Adult.** It is used to predict whether an individual's income exceeds \$50,000 per year based on features like age, education, occupation, and marital status. Key sensitive attributes include gender (67% male, 33% female) and race (85% white, 15% non-white). The dataset has a class imbalance, with 76% earning less than \$50K and 24% earning more. Due to its demographic information, the dataset is frequently used in fairness research to examine potential biases and disparate impacts across different groups. Dataset details are provided below.

2) **Bank.** It is used to predict whether a client will subscribe to a term deposit based on characteristics such as age, job, education, and previous marketing interactions. Key sensitive attributes include age, marital status (60% married, 28% single), and education level. The dataset is imbalanced, with 88% of clients not subscribing to the deposit. Due to the inclusion of demographic and socio-economic factors, this dataset is often analyzed in fairness studies to explore potential biases and unequal treatment across different groups.

3) **Default.** It is used to predict whether a client will default on a credit card payment based on features like balance, income, and past payment history. Key sensitive attributes include gender and age, which are important for fairness analysis. The dataset has an imbalance, with a higher proportion

of non-default cases. This dataset is often studied in fairness research to identify and mitigate potential biases, ensuring that credit decisions do not disproportionately disadvantage certain demographic groups.

4) **Law.** It is used to predict whether a law school applicant will pass the bar exam based on features like LSAT scores, undergraduate GPA, and law school rank. It includes sensitive attributes such as race and gender, making it a valuable resource for fairness research. The dataset shows disparities in bar passage rates across different racial groups, highlighting the need to analyze and address potential biases in admissions and evaluation processes.

5) **KDD.** It is used to predict whether an individual's income exceeds \$50,000 per year based on features such as age, education, and occupation. Key sensitive attributes include gender and race, making it a significant resource for fairness research. The dataset is imbalanced, with a larger proportion of individuals earning less than \$50K. It is commonly used to study and address biases in predictive models, ensuring equitable outcomes across different demographic groups. The prediction task is the same as the Adult dataset. However, the differences between the two datasets are: "the goal field was drawn from the total person income field rather than the adjusted gross income and may, therefore, behave differently than the original adult goal field."

Baselines. We consider FedAvg [42], Agnostic-Fair [28], FCFL [29], FairFed [17], FedFB [30], FairTrade [12] for comparison.

Settings. Evaluation is performed for: (1) comparison with baseline methods; (2) generalisability with (a) fairness-constrained optimization with different metrics and (b) data distributions: random (R) and attribute-based ($Attr$), with a varying number of clients (C); (3) non-iid distribution; (4) (a) sensitivity to number of important weights selected and (b) β_1 and β_2 ; (5) computation and communication overheads.

Parameters. Unless specified, we consider the number of clients as 3, total global communication rounds as 50, local as 15, and data distribution as random. R3C represents a random data distribution with 3 clients. The model used is similar to FairTrade [12]. This is a feedforward neural network comprising three fully connected layers with hidden dimensions of 64 and 32, respectively. Each hidden layer is followed by a ReLU activation, and a final sigmoid activation is applied to produce a scalar output suitable for binary classification. The learning rate is set as 0.001, and the regularisation constraint is 100. The number of weights selected as responsible for bias is set as 10% of the total weights. β_1 and β_2 are initialised as 1. We ran the experiments 10 times each and reported the average result. We used the implementation provided in FairTrade as baseline to implement FairWeight. The code can be referred to here¹.

A. Results and Discussion

Comparison. FairWeight is compared with five baselines under the same experimental setup of FairTrade [12]. Table II

reports the result for accuracy (Acc), balanced accuracy (BA), and demographic parity (DP) among 3 clients with random data distribution. The local fairness-constrained optimization is based on DP. The reported results are averaged over 10 runs with insignificant standard deviation in order of 10^{-3} . All the baselines other than FedAvg and FairTrade have a balanced accuracy of nearly 50%. FedAvg has high balanced accuracy, but at the cost of poor DP. FairWeight performs significantly better in achieving a balance between BA and DP than FairTrade, for 3 datasets. For the law dataset, it achieves a 10% higher BA than FairTrade. However, it has a 0.02% lower BA for Adult and KDD datasets. This is because (1) the Adult dataset causes poor global model accuracy itself, and (2) KDD has an almost balanced distribution among the two groups of sensitive attributes and carries very little bias. Some responsible weights identified by WI might be causing both bias and misclassification in the scenario of class imbalance (e.g., KDD dataset). This leaves an opportunity for enhanced model training and scoring separately for fairness and imbalance.

Generalisability. We only consider FairTrade for comparison with remaining settings, since our approach improves their design. FairWeight is evaluated with fairness-constrained optimization based on causal fairness (FACE). Table III reports the result for BA, DP, and FACE among 3 clients with random data distribution. FairWeight achieves high BA and low DP, which makes it agnostic for fairness metric. FairTrade achieves high BA for Adult and KDD for reasons similar to those discussed above. FairWeight is also evaluated for different data distributions with varying numbers of clients. Table IV reports the result for BA, DP, and FACE in comparison with FairTrade, over the law dataset (*we choose the law dataset, since FairTrade has a significant loss of BA over Acc on this dataset*). In Attr3C, each client dataset has data associated with different categories of an attribute. FairWeight has better performance with both Attr3C and an increasing number of clients. Since FairTrade achieves high BA and fairness in 10 rounds and maintains it, these results are also reported over only 10 global communication rounds, to test the efficacy of FairWeight. The improvement in results over FairTrade reported in Tables II-IV also demonstrates the benefit of our design choice, which calculates the aggregation score of clients coordinate-wise, utilising both optimization and calculating the frequency of responsible weights across the local models. Further, we evaluate the impact on non-IID settings by varying the balance rate ($br = 0.25, 0.50, 0.75$), which is the ratio between one class data volume and its counterpart [43]. Table V reports the performance of FairWeight under varying degrees of non-IID data partitioning. As expected, when λ increases, FairWeight consistently improves BA and DP.

Sensitivity. Table VI demonstrates the performance of FairWeight, with a varying number of responsible weights selected for bias and misclassification. It can be seen that selecting 10-15% of the weights achieves the best fairness scores, with a small trade-off to balanced accuracy. Selecting more weights and then incentivising or penalising can cause a decrease in accuracy and an increase in bias. This is because, some important weights for the main classification task also get

¹<https://tinyurl.com/fairweight>

TABLE II: Comparison with baseline methods with R3C, in terms of Acc, BA, and DP. [Bold represents the best BA].

Dataset	FedAvg			Agnostic-Fair			FCFL			FedFB			FairFed			FairTrade			FairWeight		
	Acc	BA	DP	Acc	BA	DP	Acc	BA	DP	Acc	BA	DP	Acc	BA	DP	Acc	BA	DP	Acc	BA	DP
Adult	0.77	0.79	0.409	0.77	0.58	0.067	0.8	0.61	0.092	0.76	0.5	0.0005	0.76	0.5	0.0008	0.76	0.77	0.001	0.71	0.75	0.01
Bank	0.87	0.79	0.076	0.86	0.53	0.064	0.88	0.55	0.003	0.89	0.56	0.0006	0.88	0.56	0.03	0.88	0.78	0.019	0.85	0.83	0.011
Default	0.78	0.69	0.053	0.78	0.52	0.006	0.8	0.57	0.089	0.8	0.58	0.008	0.79	0.58	0.026	0.77	0.69	0.01	0.75	0.71	0.015
Law	0.88	0.72	0.037	0.93	0.5	0.001	0.91	0.61	0.029	0.91	0.57	0.004	0.91	0.59	0.021	0.88	0.68	0.008	0.83	0.78	0.005
KDD	0.94	0.59	0.012	0.93	0.5	0.001	0.93	0.52	0.003	0.93	0.5	0.001	0.93	0.5	0.001	0.87	0.82	0.002	0.93	0.8	0.001

TABLE III: Results with causal fairness (FACE) constrained optimization. [Bold represents the best BA for each dataset].

Dataset	FairTrade			FairWeight		
	BA	DP	FACE	BA	DP	FACE
Adult	0.7945	0.0913	-0.0094	0.721	0.01	-0.004
Bank	0.806	0.0314	0.009	0.814	0.015	-0.019
Default	0.6981	0.0227	0.0117	0.7	0.017	0.008
Law	0.7467	0.0333	0.010	0.773	0.01	0.002
KDD	0.8196	0.0558	0.011	0.8	0.04	-0.008

TABLE IV: Results with different data distribution: random (R), attribute-based (Attr) with a varying number of clients (C). [Bold represents the best BA for each dataset].

Split	FairTrade			FairWeight		
	BA	DP	FACE	BA	DP	FACE
Attr3C	0.74	0.003	0.007	0.76	0.002	0.006
R3C	0.76	0.015	0.003	0.78	0.003	-0.001
R5C	0.76	0.019	0.008	0.77	0.003	0.002
R10C	0.74	0.006	0.018	0.77	0.001	0.003
R15C	0.72	0.020	0.010	0.77	0.001	0.003

TABLE V: BA, DP and FACE for FairWeight with non-IID partitions [Bold represents the best value for each fairness metric (column)].

λ	Bank			Law		
	BA	DP	FACE	BA	DP	FACE
0.25	0.731	0.027	-0.0012	0.742	0.0022	0.0013
0.50	0.745	0.019	0.0019	0.757	0.0021	-0.0009
0.75	0.772	0.012	0.0005	0.762	0.0011	-0.0019

TABLE VI: BA, DP and FACE for FairWeight with R3C [Bold represents the best value for each fairness metric (column)].

Weights (%)	Bank			Law		
	BA	DP	FACE	BA	DP	FACE
5	0.797	0.030	-0.013	0.78	0.0025	0.0042
10	0.771	0.012	-0.006	0.768	0.0021	-0.0029
15	0.774	0.014	0.0012	0.769	0.001	-0.007
20	0.773	0.019	-0.0069	0.774	0.003	0.003
25	0.779	0.027	-0.0067	0.775	0.0029	0.0049

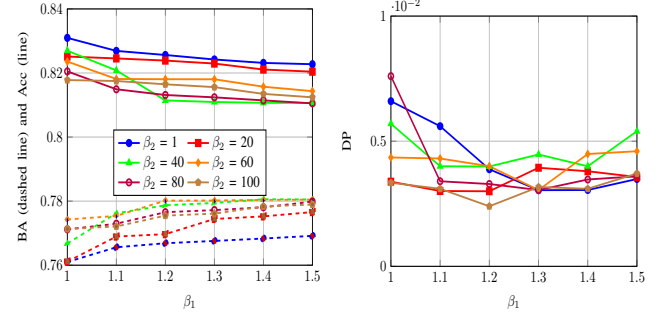


Fig. 3: Sensitivity of β_1 and β_2 to Acc and BA.

penalised, thus impacting both accuracy and fairness. Figure 3 illustrates the sensitivity of β_1 and β_2 with R3C on the law dataset. For all values of β_1 , with increasing β_2 , BA increases and Acc decreases. DP decreases first and then tends to increase with a higher value of β_1 and β_2 , because it even penalises the good weights for classification.

Computation and communication overheads. The execution time for FairTrade with 3 clients for 10 global rounds, is 3 minutes. Using SecAgg, FairWeight takes only 20 seconds(s) to execute. Integrating HE adds computation overheads, in which the time taken for each operation is as: (1) encryption: 8s, (2) addition/subtraction: 1.3s, (3) multiplication: 9s, (4) decryption: 19s. Server requires 5 additions and 5 multiplications, which adds an overhead of 51.5s to the server. Each client takes approximately 27s to encrypt and decrypt. In total, it takes 152.5s, which is still better than FairTrade. FairWeight requires only one communication between server and client, compared to two in FairFed and many in FairTrade.

Pareto efficiency analysis: Figure 4 illustrates the three-way trade-off between fairness (DP), balanced accuracy (BA), and runtime overhead (in seconds). DP and BA are reported on the Bank dataset. Each point corresponds to a fairness-aware aggregation rule, with runtime also encoded by color and explicitly annotated. The plot highlights that while methods such as FedFB and FCFL achieve very low DP, but they also have low BA and substantially higher runtime. Similarly, FedAvg and Agnostic-Fair sacrifice fairness for accuracy. In contrast, FairWeight (particularly with SecAgg) achieves an acceptable balance, lying close to the Pareto front by simultaneously delivering high balanced accuracy (0.83), low DP (0.011), and competitive runtime (20s). This demonstrates that FairWeight provides a near optimal fairness accuracy efficiency trade-off compared to existing baselines.

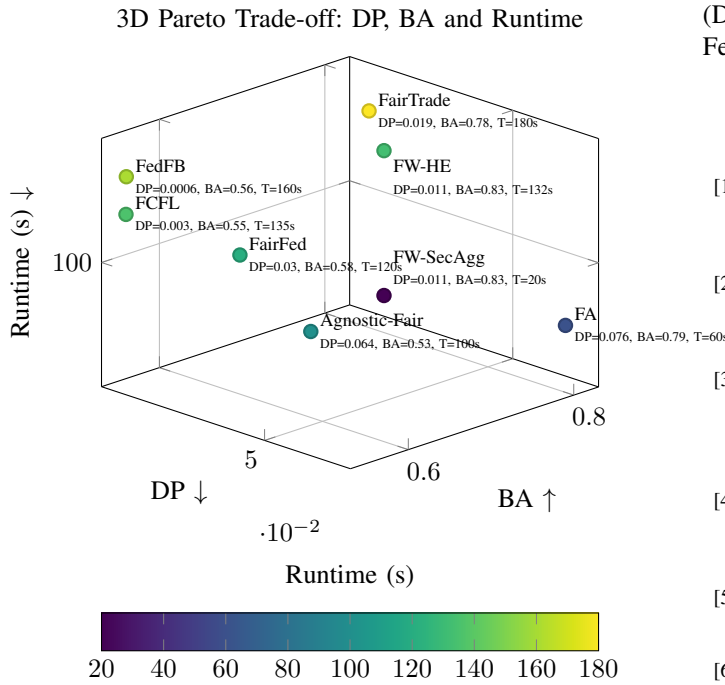


Fig. 4: 3D Pareto trade-off between fairness (DP), balanced accuracy (BA), and runtime. Each method is annotated with exact numerical values shown on a separate line for clarity.

VI. CONCLUSION

We proposed a novel fairness-aware aggregation method in FL, called FairWeight. It used fairness-constrained optimization, supporting different fairness metrics. Further, it employed a coordinate-wise aggregation weightage scheme, using model interpretability technique to find the weights causing bias and misclassification across the clients. It incentivised/penalised the weights selectively, achieving both balanced accuracy and fairness. The empirical evaluation over multiple datasets and comparison with baseline schemes demonstrated the efficacy of FairWeight. FairWeight also ensured no privacy leakage, using either SecAgg or HE. FairWeight assumes that protected attributes are available locally at the client side. While this is reasonable in many regulated domains such as healthcare or finance, future work should consider fairness-agnostic or representation-learning approaches that do not require explicit demographic information.

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REFERENCES

- [1] H. B. McMahan, E. Moore, D. Ramage, and B. A. y Arcas, "Federated learning of deep networks using model averaging," *CoRR*, vol. abs/1602.05629, 2016. [Online]. Available: <http://arxiv.org/abs/1602.05629>
- [2] P. Kairouz, H. B. McMahan, B. Avent, A. Bellet, M. Bennis, A. N. Bhagoji, K. Bonawitz, Z. Charles, G. Cormode, R. Cummings *et al.*, "Advances and open problems in federated learning," *Foundations and trends in machine learning*, vol. 14, no. 1–2, pp. 1–210, 2021.
- [3] P. P. Kulkarni, H. Kasyap, and S. Tripathy, "Dnet: an efficient privacy-preserving distributed learning framework for healthcare systems," in *Distributed Computing and Internet Technology: 17th International Conference, ICDCIT 2021, Bhubaneswar, India, January 7–10, 2021, Proceedings 17*. Springer, 2021, pp. 145–159.
- [4] Z. L. Teo, L. Jin, N. Liu, S. Li, D. Miao, X. Zhang, W. Y. Ng, T. F. Tan, D. M. Lee, K. J. Chua *et al.*, "Federated machine learning in healthcare: A systematic review on clinical applications and technical architecture," *Cell Reports Medicine*, vol. 5, no. 2, 2024.
- [5] T. Liu, Z. Wang, H. He, W. Shi, L. Lin, R. An, and C. Li, "Efficient and secure federated learning for financial applications," *Applied Sciences*, vol. 13, no. 10, p. 5877, 2023.
- [6] R. Kang, Q. Li, and H. Lu, "Federated machine learning in finance: A systematic review on technical architecture and financial applications," *Applied and Computational Engineering*, vol. 102, pp. 61–72, 2024.
- [7] H. Kasyap, U. I. Atmaca, and C. Maple, "Private fairness-aware aggregation in federated learning for financial fraud detection," in *IET Conference Proceedings CP940*, vol. 2025, no. 22. IET, 2025, pp. 182–183.
- [8] K. Maeng, H. Lu, L. Melis, J. Nguyen, M. Rabbat, and C.-J. Wu, "Towards fair federated recommendation learning: Characterizing the inter-dependence of system and data heterogeneity," in *Proceedings of the 16th ACM Conference on Recommender Systems*, 2022, pp. 156–167.
- [9] M. Bankhwal, M. Chiu, A. Bisht, R. Roberts, and A. van Heteren, "Ai for social good: Improving lives and protecting the planet," *McKinsey & Company*, 2024, accessed on 20 January 2025. [Online]. Available: <https://www.mckinsey.com/capabilities/quantumblack/our-insights/ai-for-social-good>
- [10] N. Bhutta, A. Hizmo, and D. Ringo, "How much does racial bias affect mortgage lending? evidence from human and algorithmic credit decisions," 2022.
- [11] H. Chang and R. Shokri, "Bias propagation in federated learning," *arXiv preprint arXiv:2309.02160*, 2023.
- [12] M. Badar, S. Sikdar, W. Nejdl, and M. Fisichella, "Fairtrade: Achieving pareto-optimal trade-offs between balanced accuracy and fairness in federated learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 10, 2024, pp. 10962–10970.
- [13] Y. Shi, H. Yu, and C. Leung, "Towards fairness-aware federated learning," *IEEE Transactions on Neural Networks and Learning Systems*, 2023.
- [14] S. Reddi, Z. Charles, M. Zaheer, Z. Garrett, K. Rush, J. Konečný, S. Kumar, and H. B. McMahan, "Adaptive federated optimization," *arXiv preprint arXiv:2003.00295*, 2020.
- [15] L. Lyu, X. Xu, Q. Wang, and H. Yu, "Collaborative fairness in federated learning," *Federated Learning: Privacy and Incentive*, pp. 189–204, 2020.
- [16] J. Chakraborty, S. Majumder, and T. Menzies, "Bias in machine learning software: Why? how? what to do?" in *Proceedings of the 29th ACM joint meeting on european software engineering conference and symposium on the foundations of software engineering*, 2021, pp. 429–440.
- [17] Y. H. Ezzeldin, S. Yan, C. He, E. Ferrara, and A. S. Avestimehr, "Fairfed: Enabling group fairness in federated learning," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 37, no. 6, 2023, pp. 7494–7502.
- [18] S. Kanaparthi, M. Padala, S. Damle, R. K. Sarvadevabhatla, and S. Gujar, "F3: fair and federated face attribute classification with heterogeneous data," in *Pacific-Asia Conference on Knowledge Discovery and Data Mining*. Springer, 2023, pp. 483–494.
- [19] T. Salazar, M. Fernandes, H. Araújo, and P. H. Abreu, "Fair-fate: Fair federated learning with momentum," in *International Conference on Computational Science*. Springer, 2023, pp. 524–538.

- [20] A. Agarwal, A. Beygelzimer, M. Dudík, J. Langford, and H. Wallach, "A reductions approach to fair classification," in *International conference on machine learning*. PMLR, 2018, pp. 60–69.
- [21] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H. B. McMahan, S. Patel, D. Ramage, A. Segal, and K. Seth, "Practical secure aggregation for privacy-preserving machine learning," in *proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017, pp. 1175–1191.
- [22] J. H. Cheon, A. Kim, M. Kim, and Y. Song, "Homomorphic encryption for arithmetic of approximate numbers," in *Advances in Cryptology—ASIACRYPT 2017: 23rd International Conference on the Theory and Applications of Cryptology and Information Security, Hong Kong, China, December 3–7, 2017, Proceedings, Part I 23*. Springer, 2017, pp. 409–437.
- [23] D. Manna, H. Kasyap, and S. Tripathy, "Milsa: Model interpretation based label sniffing attack in federated learning," in *International Conference on Information Systems Security*. Springer, 2022, pp. 139–154.
- [24] L. Bai, H. Hu, Q. Ye, H. Li, L. Wang, and J. Xu, "Membership inference attacks and defenses in federated learning: A survey," *ACM Computing Surveys*, vol. 57, no. 4, pp. 1–35, 2024.
- [25] H. Kasyap, U. I. Atmaca, M. Iezzi, T. Walsh, and C. Maple, "Mitigating bias: Model pruning for enhanced model fairness and efficiency," in *Frontiers in Artificial Intelligence and Applications, ECAI 2024*, vol. 392. IOS Press, 2024, pp. 995–1002.
- [26] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, "Fairness through awareness," in *Proceedings of the 3rd innovations in theoretical computer science conference*, 2012, pp. 214–226.
- [27] A. Khademi, S. Lee, D. Foley, and V. Honavar, "Fairness in algorithmic decision making: An excursion through the lens of causality," in *The World Wide Web Conference*, 2019, pp. 2907–2914.
- [28] W. Du, D. Xu, X. Wu, and H. Tong, "Fairness-aware agnostic federated learning," in *Proceedings of the 2021 SIAM International Conference on Data Mining (SDM)*. SIAM, 2021, pp. 181–189.
- [29] S. Cui, W. Pan, J. Liang, C. Zhang, and F. Wang, "Addressing algorithmic disparity and performance inconsistency in federated learning," *Advances in Neural Information Processing Systems*, vol. 34, pp. 26 091–26 102, 2021.
- [30] Y. Zeng, H. Chen, and K. Lee, "Improving fairness via federated learning," *arXiv preprint arXiv:2110.15545*, 2021.
- [31] Y. Roh, K. Lee, S. E. Whang, and C. Suh, "Fairbatch: Batch selection for model fairness," *arXiv preprint arXiv:2012.01696*, 2020.
- [32] D. Yin, Y. Chen, R. Kannan, and P. Bartlett, "Byzantine-robust distributed learning: Towards optimal statistical rates," in *International Conference on Machine Learning*. PMLR, 2018, pp. 5650–5659.
- [33] H. Kasyap and S. Tripathy, "Sine: Similarity is not enough for mitigating local model poisoning attacks in federated learning," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 5, pp. 4481–4494, 2024.
- [34] V. Valadi, X. Qiu, P. P. B. de Gusmão, N. D. Lane, and M. Alibeigi, "FedVal: Different good or different bad in federated learning," in *32nd USENIX Security Symposium (USENIX Security 23)*. Anaheim, CA: USENIX Association, Aug. 2023, pp. 6365–6380. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity23/presentation/valadi>
- [35] S. Shaham, A. Hajisafi, M. K. Quan, D. C. Nguyen, B. Krishnamachari, C. Peris, G. Ghinita, C. Shahabi, and P. N. Pathirana, "Privacy and fairness in machine learning: A survey," *IEEE Transactions on Artificial Intelligence*, 2025.
- [36] H. Zhang, X. Yuan, and S. Pan, "Unraveling privacy risks of individual fairness in graph neural networks," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*. IEEE, 2024, pp. 1712–1725.
- [37] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial Intelligence and Statistics*. PMLR, 2017, pp. 1273–1282.
- [38] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in neural information processing systems*, 2017, pp. 4765–4774.
- [39] M. Kang, L. Li, and B. Li, "Fashapley: Fast and approximated shapley based model pruning towards certifiably robust dnn," in *2023 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*. IEEE, 2023, pp. 575–592.
- [40] K. Bache and M. Lichman, "Uci machine learning repository," 2013.
- [41] L. F. Wightman, "Lsac national longitudinal bar passage study. Isac research report series." 1998.
- [42] J. Konečný, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, and D. Bacon, "Federated learning: Strategies for improving communication efficiency," *arXiv preprint arXiv:1610.05492*, 2016.
- [43] A. Li, J. Sun, B. Wang, L. Duan, S. Li, Y. Chen, and H. Li, "Lotteryfl: Empower edge intelligence with personalized and communication-efficient federated learning," in *2021 IEEE/ACM Symposium on Edge Computing (SEC)*, 2021, pp. 68–79.
- [44] C. Baritel-Ruet, "Formal security proofs of cryptographic: A necessity achieved using easycrypt," Ph.D. dissertation, Université côte d'azur, 2020.
- [45] D. Zhao, "Communication-efficient search under fully homomorphic encryption for federated machine learning," *arXiv preprint arXiv:2308.04648*, 2023.



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APPENDIX

A. Weight Importance

Algorithm 2 Weight Importance

Input: *repeats* (number of repeats), *k* (number of weights to be randomly sampled)

Output: *WI* (Weight Importance (Shapley Value))

```

1: function WEIGHTIMPORTANCE(data, weights)
2:    $n \leftarrow \text{len}(\text{weights})$ 
3:    $WI \leftarrow [0 \text{ for } i \text{ in } n]$ 
4:   for  $x, y$  in data do
5:     for  $r$  in repeats do
6:        $\text{indices} \leftarrow \text{rand}(n, n - k) \triangleright$  Randomly select
         $(n - k)$  weights to be pruned.
7:        $\text{pruned\_weights} \leftarrow \text{weights}[\text{indices}] = 0$ 
8:        $\text{pruned\_model} \leftarrow \text{Load } \text{pruned\_weights}.$ 
9:        $\text{prediction}(\hat{y}) \leftarrow \text{pruned\_model}(x)$ 
10:       $\text{loss} \leftarrow L(y, \hat{y})$ 
11:       $\text{loss.backward}()$ 
12:       $\text{prev\_index} = 0$ 
13:       $\text{index} = 0$ 
14:      for  $\text{param}$  in  $\text{pruned\_model.parameters}()$ 
15:      do
16:         $\text{prev\_index} = \text{index}$ 
17:         $\text{index} += \text{len}(\text{param.flatten}())$ 
18:        if  $\text{param.grad} \neq \text{None}$  then
19:           $w \leftarrow \text{weights}[\text{prev\_index} : \text{index}]$ 
20:           $v \leftarrow \|\text{param.grad.flatten()} * w\|$ 
21:           $WI[\text{prev\_index} : \text{index}] += v$ 
22:        end if
23:      end for
24:    end for
25:  return WI
26: end function

```

The function `WeightImportance` computes the significance of the weight parameters in a model f [25]. As shown in Algorithm 2, lines 6–8: (1) a random subset of weights is selected; (2) these weights are pruned; and (3) a pruned model is reconstructed using the remaining weights. In lines 9–11, the pruned model is evaluated, and gradients are obtained through backpropagation. Finally, in lines 14–24, the importance of each weight is computed as the norm of the product of its gradient and value, provided that the gradient is not `None`.

B. Selecting β_1 and β_2

β_1 and β_2 can be adjusted (increased or decreased), based on whether the desired conditions are met, in every iteration, as demonstrated in Algorithm 3.

C. Homomorphic Encryption (HE)

In our work, we employ the CKKS scheme [22], a practical FHE scheme that supports arithmetic on approximate real

Algorithm 3 Optimize β_1 and β_2

Input: Initialise $\beta_1, \beta_2 = 1$; $\text{bal_acc}_0 = 0$; $\text{fair_loss}_0 = 1$;
Initialise λ_1 in $(0, \frac{1}{n-1})$ and $\lambda_2 > 1$.

Output: β_1 and β_2 for next iteration.

```

1: for each round  $t$  in  $1, 2, \dots, T$  do
2:   if  $\text{bal\_acc}_{t-1} > \text{bal\_acc}_t$  or  $\text{fair\_loss}_{t-1} < \text{fair\_loss}$ 
3:   then
4:     if  $\beta_1 + \lambda_1 < \frac{n}{n-1}$  then
5:        $\beta_1 = \beta_1 + \lambda_1$ 
6:     end if
7:      $\beta_2 = \beta_2 + \lambda_2$ 
8:   else
9:     if  $\beta_1 - \lambda_1 \geq 1$  then
10:       $\beta_1 = \beta_1 - \lambda_1$ 
11:    end if
12:    if  $\beta_2 - \lambda_2 \geq 1$  then
13:       $\beta_2 = \beta_2 - \lambda_2$ 
14:    end if
15:  end for

```

numbers, making it suitable for floating-point operations. The security of CKKS is based in the Ring Learning With Errors (RLWE) assumption. To ensure privacy, CKKS introduces noise during encryption. Furthermore, it carefully manages numerical precision through operations like encoding and rescaling, to control accumulated approximation errors. The operations are described in detail below:

Homomorphic Addition (HEAdd): Let $ct = (c_0, c_1)$ and $ct' = (c'_0, c'_1)$ be two ciphertexts encrypting plaintexts m and m' , respectively. Their homomorphic sum is computed as:

$$ct_{\text{add}} = ct + ct' = (c_0 + c'_0, c_1 + c'_1). \quad (16)$$

This results in a ciphertext that encrypts $m + m'$, preserving the approximate relationship $c_0 + c_1 \cdot s \approx m + m' \pmod{Q}$.

Homomorphic Multiplication (HEMult): To multiply two ciphertexts $ct = (c_0, c_1)$ and $ct' = (c'_0, c'_1)$, a tensor (component-wise) multiplication is first performed:

$$(c_{\times,0}, c_{\times,1}, c_{\times,2}) = (c_0 \cdot c'_0, c_0 \cdot c'_1 + c_1 \cdot c'_0, c_1 \cdot c'_1). \quad (17)$$

This increases the ciphertext degree from 2 to 3. A relinearization step is then applied to reduce the ciphertext back to a manageable size (degree 2) using key-switching techniques. The result approximates the product $m \cdot m'$:

$$c_{\times,0} + c_{\times,1} \cdot s + c_{\times,2} \cdot s^2 \approx m \cdot m' \pmod{Q}. \quad (18)$$

Due to the multiplication, the scale of the ciphertext squares and must be adjusted using a rescaling operation to prevent error accumulation.

Multiplication by Constant: If a ciphertext $ct = (c_0, c_1)$ is multiplied by a scalar plaintext value m , the operation yields:

$$(c_0 \cdot m, c_1 \cdot m) \pmod{Q}, \quad (19)$$

assuming m is encoded into plaintext form before multiplication.

Theorem 1 (Semantic Security of the Encryption Scheme). *The public-key encryption scheme is semantically secure under a chosen plaintext attack, and ensures data confidentiality.*

Proof. Security can be defined in terms of computational indistinguishability according to assumptions made about an adversary's capabilities [44]. If the encryption scheme ensures indistinguishability under chosen plaintext attack, *i.e.*, IND-CPA, it can guarantee data confidentiality.

We start with the adversary ($\mathcal{A}$) choosing a pair of plaintext messages. The aim of $\mathcal{A}$ is to learn from the ciphertext. It is also assumed that the input messages (local model updates) belong to a fixed input space. The advantage of $\mathcal{A}$ under IND-CPA, against encryption scheme E is defined as below:

$$\text{Adv}_E^{\text{IND-CPA}}(\mathcal{A}) := \left| \Pr [r \leftarrow \text{IND-CPA}(\mathcal{A}, E) \cdot O() : r = 1] - \frac{1}{2} \right|,$$

where, an algorithm O behaves the same as the encryption scheme, and interacts with $\mathcal{A}$ under the capability IND-CPA (where the adversary is allowed to make encryption queries, but no decryption queries). O initializes the key pair (sk, pk) . $\mathcal{A}$ submits a pair of messages (m_0, m_1) to O , for executing the encryption oracle (E) over it. O randomly samples a bit $r \in \{0, 1\}$, and executes E to get c_r (corresponding to m_r), then responds c_r to $\mathcal{A}$. $\mathcal{A}$ could ask multiple encryption queries (m_0, m_1) to O . On receiving c_b from O , if $\mathcal{A}$ guesses bit r' . And $r = r'$ with probability greater than $\frac{1}{2}$, it means $\mathcal{A}$ has a non-negligible advantage, and the encryption scheme (E) is not secure.

By contradiction, it can be proved that the assumption of an adversary $\mathcal{A}$ with the ability to match the plaintext (from its own knowledge) to the received ciphertext in order to break IND-CPA security definition, is false [45].

Thus, E is secure considering the semi-honest threat model.

□